Enhanced Carpi's Test for Morphisms (preliminary observations)

Veikko Keränen, 28 July 2005
Rovaniemi Polytechnic
School of Technology
veikko.keranen@ramk.fi

The aim is to show that the global structure of long abelian squares in $h(w)$:s can be hidden in – or be visible from – the factors of short image words. Only 1 or 2 blocks may be enough instead of 3 in Carpi's method from 1993 (which has been designed for testing whether a (completely) given morphism is a-2-free or even a-n-free).

The suggested new approach should be useful when one tries to find new a-2-free morphisms, or when showing that such a morphism cannot exist for given A=$h(a)$, B=$h(b)$ (or for more given A, B, C, ... blocks – depending on the size of the alphabet).

However, in the case of the endomorphism g_{98}, the factors $u_1 u_2$ and $u_1' u_2'$ (the twins) inside ***ADA*** (+ cyclic permutations) cannot be reduced to the case of less than 3 blocks (do note that the related problematic factors can be avoided in D0L or DT0L sequences).

Especially note that in the constructs/visualisations below one can always compute the short structures backwords from the obtained longer ones, i.e., from the $X|Y|Z$ case of Carpi.

Moreover, the problematic factors $u_1 u_2$ come in pairs – or in triplets.

Parikh vectors of factors and prefixes (of image words of letters)

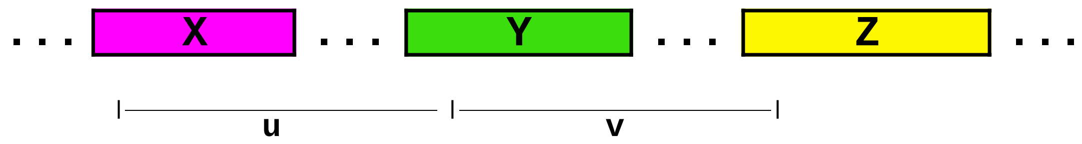
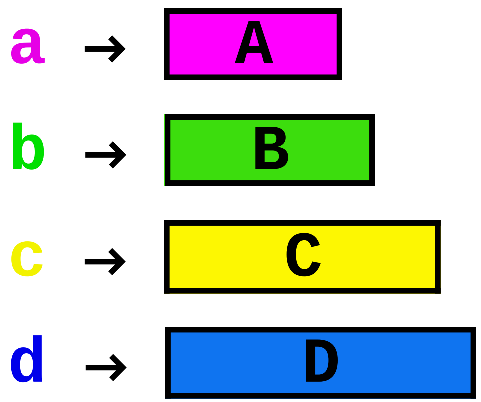
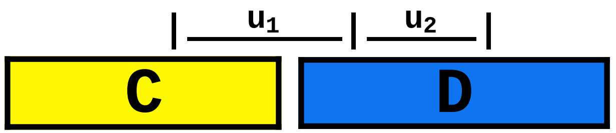


Image words of letters:

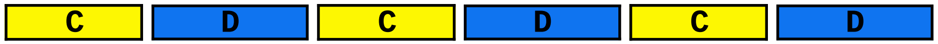


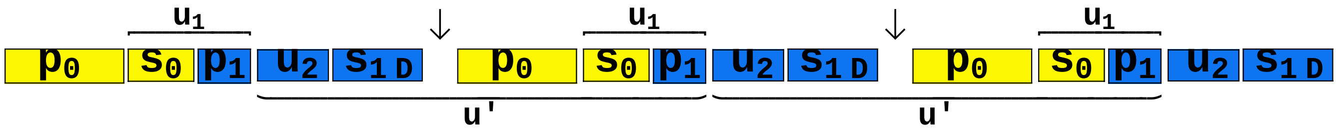
Example 1. $\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$ **inside the block** ***CD***

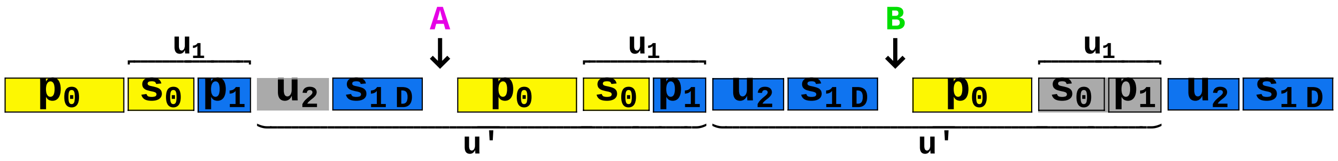
$$\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$$



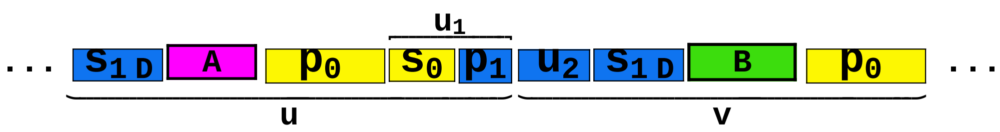
We will see that this morphism g cannot be a-2-free.







Cut away u_2 from the left and u_1 from the right. (another approach: add u_1 to the left and u_2 to the right):



where $\psi(u_2) - \psi(u_1) = \psi(\boldsymbol{A}) - \psi(\boldsymbol{B})$.

Thus

$$
\begin{aligned}
\psi(v) - \psi(u) \;&=\; \psi(u_2\, s_{1\,\mathrm{D}}\, \boldsymbol{B}\, p_0) - \psi(s_{1\,\mathrm{D}}\, \boldsymbol{A}\, p_0\, u_1) \\
&=\; \psi(u_2) - \psi(u_1) - (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) \\
&=\; (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) - (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) \\
&=\; \overline{0},
\end{aligned}
$$

i.e., uv is an abelian square.

Now $DACDBC = g(dacdbc)$ contains an abelian square though $dacdbc$ is a-2-free. Thus the morphism g is not a-2-free.

The same example in terms of Carpi's method from 1993 (using prefixes of C and D)

Proposition 1 (Carpi [1]). Given an integer $n \geq 2$, two alphabets Σ and Δ, and a morphism $h\colon \Sigma^* \to \Delta^*$, let us denote by G_h the subgroup of $\mathbb{Z}^\Delta$ generated by $\psi(h(\Sigma))$. Then, the morphism h is an abelian nth power-free (a-n-free) provided that the following conditions are satisfied:

(i) $h(w)$ is a-n-free for every a-n-free word $w \in \Sigma^*$ of length 2,

(ii) h is commutatively bijective (i.e., the set $\psi(h(\Sigma))$ is linearly independent),

(iii) for all $a_i \in \Sigma$, $p_i \in \mathrm{PREF}(\, h(a_i)\,) - \{h(a_i)\}$ $(0 \leq i \leq n)$ such that

$$\psi(p_{j+1}) - 2\,\psi(p_j) + \psi(p_{j-1}) \in G_h, \qquad j = 1, 2, \ldots, n-1,$$

there exists integers $\delta_i \in \{0,1\}$ such that

$$\psi(p_{j+1}) - 2\,\psi(p_j) + \psi(p_{j-1}) = \delta_{j+1}\,\psi(h(a_{j+1})) - 2\,\delta_j\,\psi(h(a_j)) + \delta_{j-1}\,\psi(h(a_{j-1})), \quad j = 1, 2, \ldots, n-1.$$

If $\mathrm{Card}(\Sigma) \geq 6$, then an a-$n$-free morphism h always satisfies conditions (i)–(iii). □

- ***Arturo Carpi: Investigations into g85 and beyond***

- In our case, let $\{\Delta_a, \Delta_b, \Delta_c, \Delta_d\} = \psi(p_{j+1}) - 2\,\psi(p_j) + \psi(p_{j-1})$.
  For every combination of prefixes, we solve the group of linear equations

$$x\,\#_a[A] + y\,\#_a[B] + z\,\#_a[C] + t\,\#_a[D] == \Delta_a;$$
$$x\,\#_b[A] + y\,\#_b[B] + z\,\#_b[C] + t\,\#_b[D] == \Delta_b;$$
$$x\,\#_c[A] + y\,\#_c[B] + z\,\#_c[C] + t\,\#_c[D] == \Delta_c;$$
$$x\,\#_d[A] + y\,\#_d[B] + z\,\#_d[C] + t\,\#_d[D] == \Delta_d;$$

for x, y, z and t.

- If the solution x, y, z and t consists of integers, then we have a problematic structure unless the linear combination
  $x\,\psi(A) + y\,\psi(B) + z\,\psi(C) + t\,\psi(D)$ is equal to
  $\delta_{j+1}\,\psi(g(a_{j+1})) - 2\,\delta_j\,\psi(g(a_j)) + \delta_{j-1}\,\psi(g(a_{j-1}))$
  for some δ_i:s in $\{0,1\}$.

Consider ***DACDBC*** in the form of the triplet

$$\mathbf{D} \,|\, \mathbf{D} \,|\, \mathbf{C} = \boxed{p_1\, u_2}\ \underbrace{s_{1\,D}\ \Big|\ \boxed{p_1}}\ \underbrace{u_2\, s_{1\,D}\ \Big|\ \boxed{p_0}}\ s_0$$

and when checking through all the triplets of prefixes,

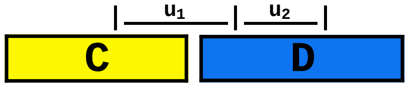
we find that

$$\psi(\boldsymbol{p_1}\,\boldsymbol{u_2}) + \psi(\boldsymbol{p_0}) - 2\psi(\boldsymbol{p_1})$$

$$\begin{aligned}
&= (\psi(\boldsymbol{D}) - \psi(\boldsymbol{s}_{1\,\mathbf{D}})) + \psi(p_0) - \psi(\boldsymbol{p}_1) - \psi(\boldsymbol{p}_1)\\
&= \psi(\boldsymbol{D}) + \psi(p_0) - (\psi(\boldsymbol{D}) - \psi(\boldsymbol{u}_2\boldsymbol{s}_{1\,\mathbf{D}})) - (\psi(\boldsymbol{s}_{1\,\mathbf{D}}) + \psi(\boldsymbol{p}_1))\\
&= \psi(p_0) + \psi(\boldsymbol{u}_2\boldsymbol{s}_{1\,\mathbf{D}}) - (\psi(\boldsymbol{s}_{1\,\mathbf{D}}) + \psi(\boldsymbol{p}_1))\\
&= \psi(\boldsymbol{u}_2\boldsymbol{s}_{1\,\mathbf{D}}\,p_0) - \psi(\boldsymbol{s}_{1\,\mathbf{D}}\,\boldsymbol{p}_1)\\
&= (\psi(v) + \psi(\boldsymbol{B})) - (\psi(u) + \psi(\boldsymbol{A}) + \psi(C))\\
&= (\psi(v) - \psi(u)) + \psi(\boldsymbol{B}) - \psi(\boldsymbol{A}) - \psi(C) \quad (uv \;\text{ a–square})\\
&= \psi(\boldsymbol{B}) - \psi(\boldsymbol{A}) - \psi(C) \quad (\in\; G_g) \quad (\, \psi(u_2) - \psi(u_1) \;= \psi(\boldsymbol{A}) - \psi(\boldsymbol{B})\,)\\
&= \psi(u_1) - \psi(u_2) - \psi(C)
\end{aligned}$$

G_g is the subgroup of $\mathbb{Z}^{\Sigma_4}$ generated by $\psi(g(\Sigma_4))$.

Observation. From $\psi(\boldsymbol{p}_1\,\boldsymbol{u}_2) + \psi(p_0) - 2\psi(\boldsymbol{p}_1) = \psi(\boldsymbol{B}) - \psi(\boldsymbol{A}) - \psi(C) = \psi(u_1) - \psi(u_2) - \psi(C)$ one can also compute backwords the structure



In this sense, the global structure of long abelian squares can be hidden in – or be visible from – the factors of short image words (only 1 or 2 blocks may be enough instead of 3 in Carpi's method from 1993).

In our case, assuming that the set $\psi(g(\Sigma_4))$ is linearly independent,

$$\begin{aligned}
&\psi(\boldsymbol{p}_1\,\boldsymbol{u}_2) - 2\,\psi(\boldsymbol{p}_1) + \psi(p_0) \;=\; \psi(\boldsymbol{B}) - \psi(\boldsymbol{A}) - \psi(C)\\
&\neq \delta_{j+1}\,\psi(g(a_{j+1})) - 2\,\delta_j\,\psi(g(a_j)) + \delta_{j-1}\,\psi(g(a_{j-1}))\\
&= \delta_{j+1}\,\psi(C) - 2\,\delta_j\,\psi(\boldsymbol{D}) + \delta_{j-1}\,\psi(\boldsymbol{D}) \quad \text{for all} \quad \delta_i \in \{0, 1\}.
\end{aligned}$$

This means by Proposition 1 (Carpi[1]) that the morphism g is not a-2-free.

Task (or conjecture). Formalize the *equivalence* of Proposition 1 (iii) (checking prefixes) and the approach of checking "differences" of short factors inside blocks. Pay attention to the fact that the problematic factors $u_1 u_2$ come in pairs (twins) – or in triplets.

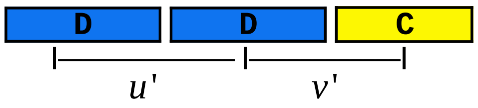
However, in the case of g_{98}, the twins inside ***ADA*** (+ cyclic permutations) cannot be reduced to the case of less than 3 blocks (but that these problematic factors can be avoided in D0L or DT0L sequences). See Example 5.

Example 2. A harmless case of *DDC*

Let $\psi(p_{j+1}) - 2\,\psi(p_j) + \psi(p_{j-1})$
$= \delta_{j+1}\,\psi(g(a_{j+1})) - 2\,\delta_j\,\psi(g(a_j)) + \delta_{j-1}\,\psi(g(a_{j-1}))$ for some $\delta_i \in \{0, 1\}$.

Say we have $\delta_{j+1} = \delta_j = \delta_{j-1} = 1$ and

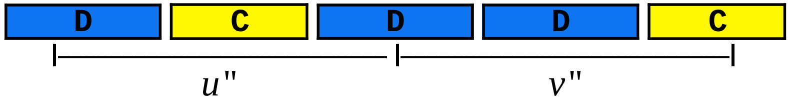
$$\psi(p_2) - 2\,\psi(\boldsymbol{p_1}) + \psi(\boldsymbol{p_0})$$
$$= \delta_{j+1}\,\psi(C) - 2\,\delta_j\,\psi(\boldsymbol{D}) + \delta_{j-1}\,\psi(\boldsymbol{D})$$
$$= \psi(C) - \psi(\boldsymbol{D})$$



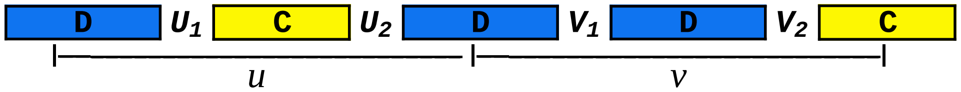
$\psi(v') - \psi(u')$
$= \psi(s_1\, p_2) - \psi(s_0\, p_1)$

$$= \psi(s_1) + \psi(p_2) - \psi(s_0) - \psi(p_1)$$
$$= (\psi(Y) - \psi(p_1)) + \psi(p_2) - (\psi(X) - \psi(p_0)) - \psi(p_1)$$
$$= \psi(Y) - \psi(X) + (\psi(p_0) + \psi(p_2) - 2\,\psi(p_1)) \quad \text{(here } X = Y = D\text{)}$$

We conclude that the abelian square in question has to occur as follows:



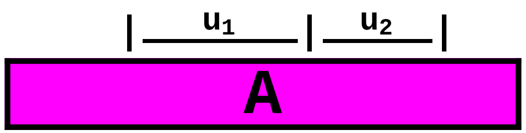
However, now there is no problem in any of the cases in



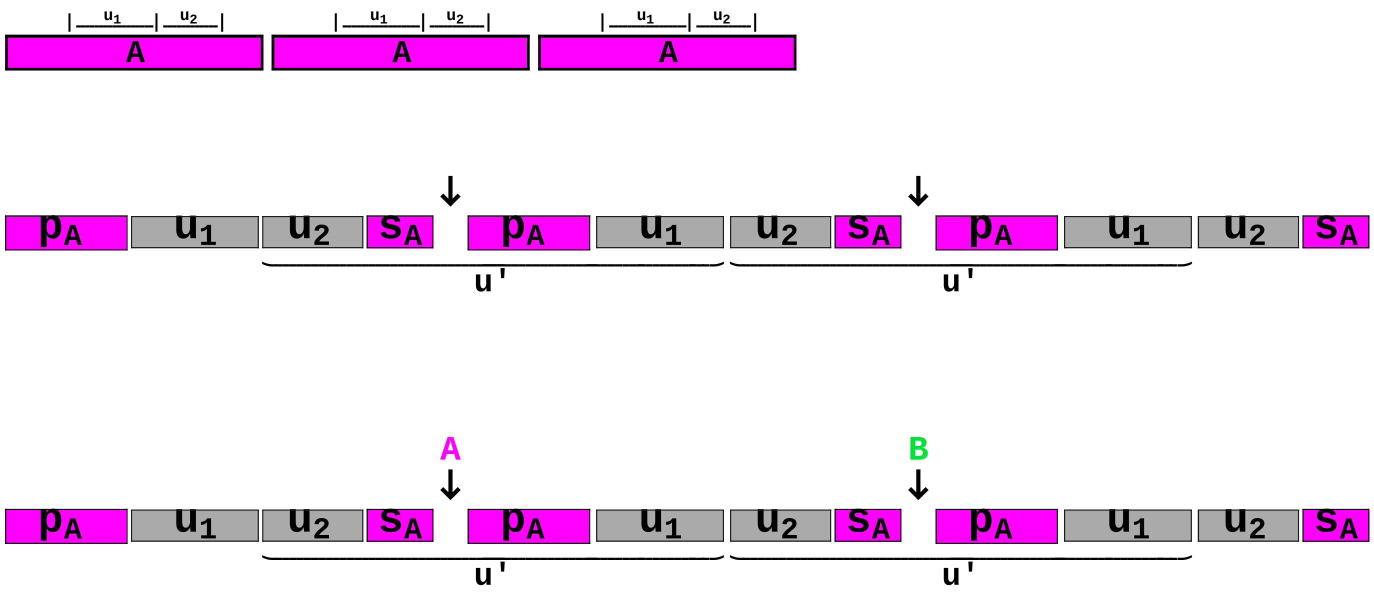
where $\psi(U_1 U_2) = \psi(V_1 V_2)$. This is because the underlying word $d\, u_1\, c\, u_2\, d\, v_1\, d\, v_2\, c$ will always contain an abelian square $(u_1\, c\, u_2\, d\, v_1\, d\, v_2\, c)$.

Example 3. $\boldsymbol{\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)}$ **inside** $\boldsymbol{A}$

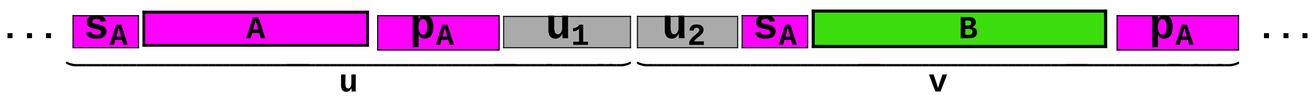
$$\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$$



We will see that this morphism g cannot be a-2-free.



Cut away u_2 from the left and u_1 from the right. (another approach: add u_1 to the left and u_2 to the right):



where $\psi(u_2) - \psi(u_1) = \psi(\boldsymbol{A}) - \psi(\boldsymbol{B})$.

Thus

$$\psi(v) - \psi(u) \;=\; \psi(u_2\, s_A\, \boldsymbol{B}\, p_A) - \psi(s_A\, \boldsymbol{A}\, p_A\, u_1)$$

$$
\begin{aligned}
&= \psi(u_2) - \psi(u_1) - (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) \\
&= (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) - (\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) \\
&= 0,
\end{aligned}
$$

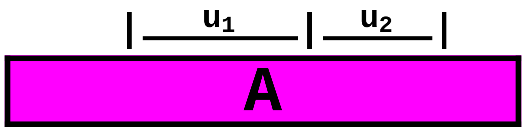
i.e., uv is an abelian square.

Now $AAABA$ can be expanded to $ACADACBDA$ (insert C and D to both sides of the middle A). Furthermore, $ACADACBDA = g(acadacbda)$ contains an abelian square though $acadacbda$ is a-2-free. Thus the morphism g is not a-2-free.

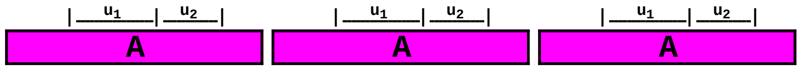
Remark. $|s_A \boldsymbol{A}\, p_A| + |u_1 u_2| = 2|A|$.
Note that both of the words (factors) $s_A \boldsymbol{A}\, p_A$ and $u_1 u_2$ are of the form $w_1 w_2$ with $\psi(w_2) - \psi(w_1) = \psi(\boldsymbol{A}) - \psi(\boldsymbol{B})$.
The words $s_A \boldsymbol{A}\, p_A$ and $u_1 u_2$ always come in pairs unless $u_1 u_2 = A$.

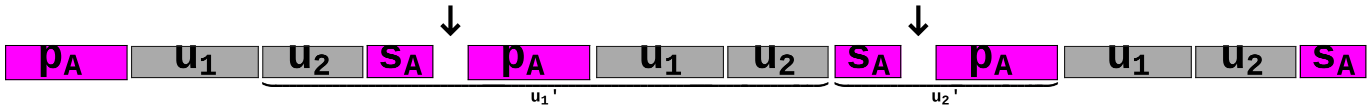
Example 4. $\psi(u_1) + 2\psi(u_2) = \psi(A) - \psi(B)$ inside A

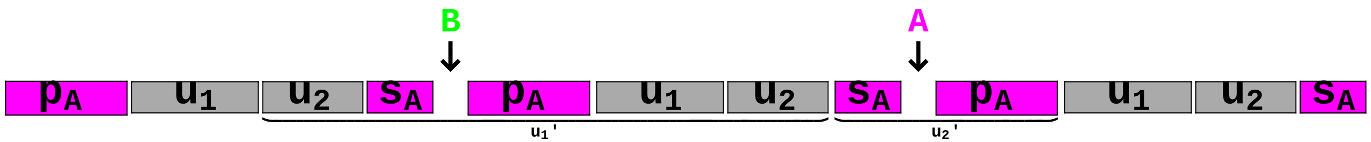
Suppose $\psi(u_1) + 2\psi(u_2) = \psi(\boldsymbol{A}) - \psi(\boldsymbol{B})$ with factors u_1 and u_2 as follows:



We will see that this morphism g cannot be a-2-free.



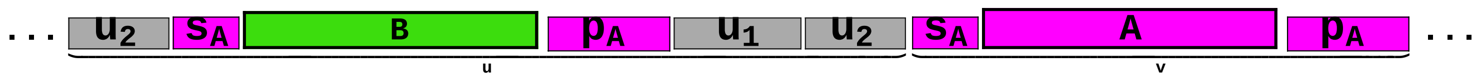




Here $|u_1'| \geq |A|$ & $|u_2'| \leq |A|$ and $\psi(u_2') - \psi(u_1') = -(\psi(u_1) + 2\psi(u_2)) = -(\psi(A) - \psi(B))$.

There would have been another option to select u_1' and u_2': just by adding u_1 to the left and right.

Now



where $\psi(u_1) + 2\psi(u_2) = \psi(A) - \psi(B)$.

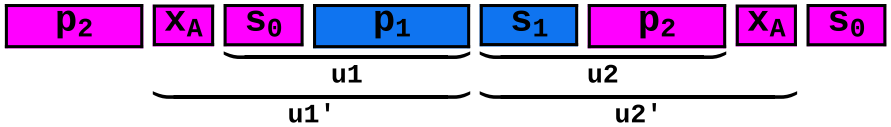
Thus

$$\begin{aligned}\psi(v)-\psi(u) &= \psi(s_A\, \boldsymbol{A}\, p_A) - \psi(u_2\, s_A\, \boldsymbol{B}\, p_A\, u_1\, u_2) \\ &= -(\psi(u_1) + 2\,\psi(u_2)) + \psi(\boldsymbol{A}) - \psi(\boldsymbol{B}) \\ &= \underline{-}(\psi(\boldsymbol{A}) - \psi(\boldsymbol{B})) + \psi(\boldsymbol{A}) - \psi(\boldsymbol{B}) \\ &= \overline{0},\end{aligned}$$

i.e., uv is an abelian square.

Now $ABAAA$ can be expanded to $ACBDACADA$ (insert C and D to both sides of the middle A). Furthermore, $ACBDACADA = g(acbdacada)$ contains an abelian square though $acbdacada$ is a-2-free. Thus the morphism g is not a-2-free.

Example 5. The case of g_{98}: $\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$ as twins inside *ADA* – in this case 3 blocks cannot be reduced.



Note that here: {A = ga, B = gb, C = gc, D = gd},
whereas {A = g98a, D = g98b, C = g98c, B = g98d})

```
{g98a, g98b, g98c, g98d}
```

```
{abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca,
 dabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcadacbcacbdbcabadbabcbdbcbacbcdcacbabd,
 cdacabadabacbdbadbdcdadbacadcdbdcdacdbcbabcbdbadbdcdbdadcadabdcdbabdbacabdadcadabacabadbabcbdbadac,
 bcdbdadcdadbacadcacbcdcadbdcbcacbcdbcabadabacadcacbcacdcbdcdacbcadacadbdacdcbdcdadbdadcadabacadcdb}
```

```
{ga, gb, gc, gd}
```

```
{abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca,
 bcdbdadcdadbacadcacbcdcadbdcbcacbcdbcabadabacadcacbcacdcbdcdacbcadacadbdacdcbdcdadbdadcadabacadcdb,
 cdacabadabacbdbadbdcdadbacadcdbdcdacdbcbabcbdbadbdcdbdadcadabdcdbabdbacabdadcadabacabadbabcbdbadac,
 dabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcadacbcacbdbcabadbabcbdbcbacbcdcacbabd}
```

```
s /@ {ga, gb, gc, gd}
```

```
{18 a + 28 b + 27 c + 25 d, 25 a + 18 b + 28 c + 27 d, 27 a + 25 b + 18 c + 28 d, 28 a + 27 b + 25 c + 18 d}
```

```
s[ga] - s[gb]
```

```
-7 a + 10 b - c - 2 d
```

```
? s
```

```
Global`s

s[x_List] := Plus @@ x

s[x_String] := Plus @@ Characters[x]
```

```
(* Clear[s]; s[ x_List ] := Apply[Plus,x]; *)

Clear[myDiff];
myDiff[x_String] :=
 Module[{pattString = {}},
  Characters[x] /. {pp___, u1__, u2__, ss___} :→ q /; If[s[{u2}] - s[{u1}] === s[ga] - s[gb],
      pattString = Append[pattString, StringJoin[pp, "|", u1, "|", u2, "|", ss]]]; pattString]

myDiff[ga] // Timing

{36.253 Second, {}}

myDiff[ga <> gd] // Timing

{323.324 Second, {}}

myDiff[gd <> ga] // Timing

{324.407 Second, {}}

myDiff[ga <> gd <> ga] // Timing

{1218.26 Second, {abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdb|
    dcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
    cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc|
    dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca, abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdc|
    dbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
    cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdb|
    abdcdbdcacdbcbacbcdcacdcbdcdadbdcbca, abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdb|
    abdcdbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
    cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdc|
    dbdcacdbcbacbcdcacdcbdcdadbdcbca, abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc|
    dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
    cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdb|
    dcacdbcbacbcdcacdcbdcdadbdcbca}}
```

The border | in u1|u2 is always at the same place:

```
{"abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdb|
  dcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
  cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc|
  dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca", "abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdc|
  dbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
  cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdb|
  abdcdbdcacdbcbacbcdcacdcbdcdadbdcbca", "abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdb|
  abdcdbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
  cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdc|
  dbdcacdbcbacbcdcacdcbdcdadbdcbca", "abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc|
  dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada|
  cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc
  dbabdcdb|dcacdbcbacbcdcacdcbdcdadbdcbca"}
```

Note the same factor *dbabdcdb* at both ends of $u_1 \mid u_2$
– *dbabdcdb* occurs at the same place in both occurrences of $A = ga$ in ADA.

The lengths of both u_1 and u_2 are 94 & 102, and 96 & 100.

This means that there are always twins $\{u_1, u_2\}$, and $\{u_1', u_2'\}$ such that

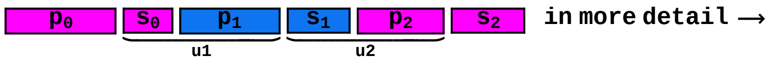
$$|u_1 u_2| + |u_1' u_2'| = 2 \cdot |AD| = 2 \cdot (98 + 98) = 2 \cdot 196,$$

where $|AD| = 196$ is the period length (ADA is a prefix of $(AD)^2$).

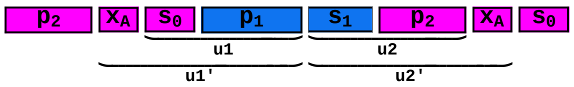
We can also say that $|u_1' u_2| = |u_1 u_2'| = |AD|$.

■ Visualisations

$\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$



$\psi(u_2) - \psi(u_1) = \psi(A) - \psi(B)$
$\psi(u_2') - \psi(u_1') = \psi(A) - \psi(B)$



Obviously $|u_1 u_2| + |u_1' u_2'| =$

$|u_1' u_2| + |u_1' u_2| = (|x_A s_0 p_1| + |s_1 p_2|) + (|s_0 p_1| + |s_1 p_2 x_A|) =$

$(|x_A s_0| + (|p_1| + |s_1|) + |p_2|) + (|s_0| + (|p_1| + |s_1|) + |p_2 x_A|) =$

$(|p_2| + |x_A s_0| + |D|) + (|p_2 x_A| + |s_0| + |D|) =$

$(|A| + |D|) + (|A| + |D|) = 2 \cdot |AD|$

In other words: $|u_1' u_2| = |u_1 u_2'| = |AD|$

```
StringLength["dcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada"]

94

{ga, gd, ga}

{"abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdb| dcacdbcbacbcdcacdcbdcdadbdcbca",
 "dabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada| cbcacbdbcabadbabcbdbcbacbcdcacbabd",
 "abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc| dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca"}
```

```
In[446]:=
StringLength /@ {{"abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdb", "dcacdbcbacbcdcacdcbdcdadbdcbca"},
  {"dabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada", "cbcacbdbcabadbabcbdbcbacbcdcacbabd"},
  {"abcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc", "dbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbca"}}

Out[446]=
{{68, 30}, {64, 34}, {60, 38}}
```

Write **ADA** = $p_0 \mid u_1 \mid u_2 \mid s_2 = p_0 \mid s_0 p_1 \mid s_1 p_2 \mid s_2$.

ADA ADA ADA = $p_0 \mid u_1 \mid u_2 \mid s_2\ p_0 \mid u_1 \mid u_2 \mid s_2\ p_0 \mid u_1 \mid u_2 \mid s_2$

A ADA A = $p_2 \mid s_2\ p_0 \mid u_1 \mid u_2 \mid s_2\ p_0 \mid s_0$

A A ADA B A = $p_2 \mid s_2 \mid$ **A** $\mid p_0 \mid u_1 \mid u_2 \mid s_2 \mid$ **B** $\mid p_0 \mid s_0$

A BAC ADA BCB A = **g (abacadabcba)** = $p_2 \mid s_2 \mid$ **BAC** $\mid p_0 \mid u_1 \mid u_2 \mid s_2 \mid$ **BCB** $\mid p_0 \mid s_0$,

where $s_2 \mid$ **BAC** $\mid p_0 \mid u_1 \mid u_2 \mid s_2 \mid$ **BCB** $\mid p_0$ **is an abelian square though abacadabcba is a - 2 - free.**

A̲CAD̲BCA̲

square starting from pos.69 and ending to pos.648 in line 160:g(acadbca)

```
StringTake[ga <> gc <> ga <> gd <> gb <> gc <> ga, {69, 648}]
```

dcacdbcbacbcdcacdcbdcdadbdcbcacdacabadabacbdbadbdcdadbacadcdbdcdacdbcbabcbdbadbdcdbdadcadabdcdbabdbacabdadcadabacabadbabc
bdbadacabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdbdcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdc
acbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcadacbcacbdbcabadbabcbdbcbacbcdcacbabdbcdbdadcdadbacadcacbcdcadbdcbcacb
cdbcabadabacadcacbcacdcbdcdacbcadacadbdacdcbdcdadbdadcadabacadcdbcdacabadabacbdbadbdcdadbacadcdbdcdacdbcbabcbdbadbdcdb
dadcadabdcdbabdbacabdadcadabacabadbabcbdbadacabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc

```
StringTake[ga <> ga <> gd <> gb <> ga, {69, 648 - 2 * 98}]
```

dcacdbcbacbcdcacdcbdcdadbdcbcaabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbcdbabdcdbdcacdbcbacbcdcacdcbdcda
dbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcadacbcacbdbcabadbabcbdbcbacbcdcacbabdbcdbdadcdadba
cadcacbcdcadbdcbcacbcdbcabadabacadcacbcacdcbdcdacbcadacadbdacdcbdcdadbdadcadabacadcdbabcacdcbcdcadbdcbdbabcbdcacbabdba
bcabdadcdadbdcbdbabdbcbacbc

```
str = StringTake[ga <> gd <> ga, {69, 648 - 2 * 98 - 2 * 98}]
```

dcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcadacbcacbdbcabadbabcbdbcbacbcd
cacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc

```
str2 = StringTake[str, {1, StringLength[str] / 2}]
str3 = StringTake[str, {StringLength[str] / 2 + 1, StringLength[str]}]
```

dcacdbcbacbcdcacdcbdcdadbdcbcadabdbcbabcbdcacbacadabacbdbadacadabdacdcbcdcacbacadacabadbabcada

cbcacbdbcabadbabcbdbcbacbcdcacbabdabcacdcbcdcadbdcbdbabcbdcacbabdbabcabdadcdadbdcbdbabdbcbacbc

```
s[str3] - s[str2]
```

$-7 a + 10 b - c - 2 d$

```
s[ga] - s[gb]
```

$-7 a + 10 b - c - 2 d$

Some Mathematica Code

In[184]:=
```
Off[General::spell]; Off[General::spell1];
```

In[208]:=
```
Clear[s]; s[x_List] := Apply[Plus, x];
Clear[root];
root[x_List] := (x /. ({p__, i___, p__, i___} | {p__, i___, p__, i___, p__}) :> {p, i})
```

In[211]:=
```
root[{C, D, C, D, C}]
```

Out[211]=
```
{C, D}
```

In[212]:=
```
root[{A, A, A}]
```

Out[212]=
```
{A}
```

In[213]:=
```
root[{D, C, D, C}]
```

Out[213]=
```
{D, C}
```

In[214]:=
```
root[{D, D, D, C}]
```

Out[214]=
```
{D, D, D, C}
```

In[434]:=

```
Clear[findTwin];
(* X = p0<>s0, Y = p1<>s1, Z = p2<>s2.  When using the code, the blocks  X, Y, Z  can be the same as well. *)

findTwin[{X_, X_, X_}, {p0_, {s0_, p1_}, {s1_, p2_}, s2_}, {vecu1_, vecu2_}, "|u1|≤|X| & |u1|≥|u2|"] :=
  Module [{newConstruct = {X, Intersection[ vecu2, {X}], Y, Intersection[ vecu1, {X}] , Z}, myRoot, lenMyRoot},
   myRoot = {X}; lenMyRoot = Length[myRoot];
   {{X}, {p2, {p2 → p1}, {p1 → p0}, s0}, {vecu1, vecu2}, "|u1|≤|" <> ToString[X] <> "| & |u1|≥|u2|"}];

findTwin[{X_, X_, X_}, {p0_, {s0_, p1_}, {s1_, p2_}, s2_}, {vecu1_, vecu2_}, "|u1|&|u2|≥|X|"] :=
  Module [{newConstruct = {X, Intersection[ vecu2, {X}], Y, Intersection[ vecu1, {X}] , Z}, myRoot, lenMyRoot},
   myRoot = {X}; lenMyRoot = Length[myRoot];
   {{X}, {p0, {p0 → p1}, {p1 → p2}, s2}, {vecu1, vecu2}, "|u1|&|u2|≥|" <> ToString[X] <> "| "}];

findTwin[{X_, X_, X_}, {p0_, {s0_, p1_}, {s1_, p2_}, s2_}, {vecu1_, vecu2_}, "|u1|≥|X| & |u2|≤|X|"] :=
  Module [{newConstruct = {X, Intersection[ vecu2, {X}], Y, Intersection[ vecu1, {X}] , Z}, myRoot, lenMyRoot},
   myRoot = {X}; lenMyRoot = Length[myRoot];
   {{X}, {p2, {p2 → p0}, {p0 → p1}, s1}, {vecu2, vecu1}, "|u1|≥|" <> ToString[X] <> "| & |u2|≤|" <> ToString[X] <> "|" <> ",  " <>
     "NOTE: " <> ToString[{p2 → p0}] <> " + 2·" <> ToString[{p0 → p1}] <> " = " <> ToString[{vecu2, vecu1}]}];

findTwin[{X_, Y_, Z_}, {p0_, {s0_, p1_}, {s1_, p2_}, s2_}, {vecu1_, vecu2_}] :=
  Module [{newConstruct = {X, Intersection[ vecu2, {X, Y, Z}], Y, Intersection[ vecu1, {X, Y, Z}] , Z}, myRoot, lenMyRoot},
   myRoot = root[Flatten[newConstruct]]; lenMyRoot = Length[myRoot];
   If[Length[myRoot] > 2, {{Last[myRoot], myRoot, First[myRoot]}, {"?"}}, {{Last[myRoot], {}, First[myRoot]},
     {p2, {s2, p1}, {p1 → p0}, s0}, {Complement[vecu1, newConstruct], Complement[vecu2, newConstruct]}}]];

(* {vecu1,vecu2} should be given in the form of multiple element sets of (reduced) blocks,
 say {{A,A,A,B,B},{C,C,D,D,D}}  in which case
    vecu2 - vecu1 = Parikh[CCDDD]- Parikh[AAABB] = 2 C̄ + 3 D̄ - 3 Ā - 2 B̄ *)

(* The rest is under construction *)
findTwin[{X_, Y_}, {p0_, {u1_}, {sX_, p1_}, s1_}, {vecu1_, vecu2_}] := {}; (* X = p0<>u1<>sX, u2 = sX<>p1 *)
findTwin[{X_, Y_}, {p0_, {s0_, p1_}, {u2_}, sY_}, {vecu1_, vecu2_}] := {}; (* Y = p1<>u2<>sY, u1 = s0<>p1 *)
findTwin[{X_}, {p0_, {u1_}, {u2_}, sX_}, {vecu1_, vecu2_}] := {};
```

```
In[321]:=
        findTwin[{D, D, C}, {p0D, {s0D, p1D}, {s1D, p2C}, s2C}, {{B}, {A, C}}]

Out[321]=
        {{C, {}, D}, {p2C, {s2C, p1D}, {p1D → p0D}, s0D}, {{B}, {A}}}

In[322]:=
        findTwin[{D, D, C}, {p0, {s0, p1}, {s1, p2}, s2}, {{B}, {A, D}}]

Out[322]=
        {{C, {D, D, D, C}, D}, {?}}

In[320]:=
        findTwin[{A, A, A}, {p0, {s0, p1}, {s1, p2}, s2}, {{B}, {A}}, "|u1|≤|X| & |u1|≥|u2|"]

Out[320]=
        {{A}, {p2, {p2 → p1}, {p1 → p0}, s0}, {{B}, {A}}, |u1|≤|A| & |u1|≥|u2|}

In[362]:=
        findTwin[{A, A, A}, {p0, {s0, p1}, {s1, p2}, s2}, {{B}, {A}}, "|u1|&|u2|≥|X|"]

Out[362]=
        {{A}, {p0, {p0 → p1}, {p1 → p2}, s2}, {{B}, {A}}, |u1|&|u2|≥|A| }

In[440]:=
        findTwin[{A, A, A}, {p0, {s0, p1}, {s1, p2}, s2}, {{B}, {A}}, "|u1|≥|X| & |u2|≤|X|"]

Out[440]=
        {{A}, {p2, {p2 → p0}, {p0 → p1}, s1}, {{A}, {B}}, |u1|≥|A| & |u2|≤|A|,  NOTE: {p2 -> p0} + 2·{p0 -> p1} = {{A}, {B}}}

In[29]:=  Intersection[ {C, C, D, D}, {D, C, D, C}]

Out[29]=  {C, D}

In[41]:=  Complement[ {D, D, C}, {C, C, D}]

Out[41]=  {}
```

In[253]:=

```
Complement[{D, C}, {C}]
```

Out[253]=

```
{D}
```

```
Clear[s]; s[ x_List ] := Apply[Plus,x];

Clear[periods];
periods[x_List]:=
(x /. {pref__,pref__,___} :> {pref})
```

In[148]:=

```
periods[{C, D, C, D, C, D}]
```

Out[148]=

```
{C, D, C, D, C, D}
```

In[141]:=

```
Clear[s]; s[ x_List ] := Apply[Plus,x];

Clear[periods];
periods[x_List]:=
(x /. {(pref__)..} :> pref)
```

In[144]:=

```
periods[{C, C, C, C, C}]
```

Out[144]=

```
C
```

```
Clear[s]; s[ x_List ] := Apply[Plus,x];

Clear[abelianPeriodAtPref];
abelianPeriodAtPref[x_List]:=
Module[{pattString = {}},
 x /. {pref__,infix__,___} :> q /;
  If[ s[{pref}]===s[{infix}],
   pattString = Append[pattString,{pref}]];pattString]
```

```
abelianPeriodAtPref[{D, C, C, C, D, C, D, C}]
```

Out[42]= {{D, C, C}}

In[441]:=

```
Clear[fff];
fff[XXX_, XXX] := XXX
```

In[444]:=

```
fff[A, XXX]
```

Out[444]=

```
A
```